

Counting Rule.

① Multiplication Rule

② Permutation (order matters)

$$(i) {}^n P_r = \frac{n!}{(n-r)!}$$

$$(ii) {}^n P_{n_1 \cdot n_2 \cdot n_3 \dots n_k} = \frac{n!}{n_1! \cdot n_2! \cdot n_3! \dots n_k!}$$

(iii) in a circle $(n-1)!$

③ Combination ${}^n C_r = \frac{n!}{r!(n-r)!}$ (any order)

$$\bar{A} = S - A$$

$$A \cup \bar{A} = S$$

$A \cup B$, A or B

$A \cap B$, A and B

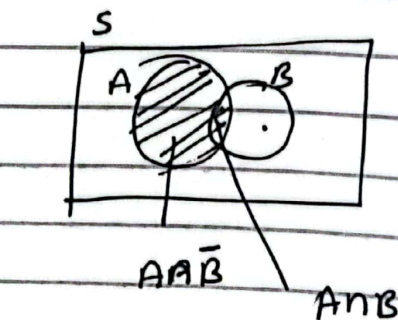
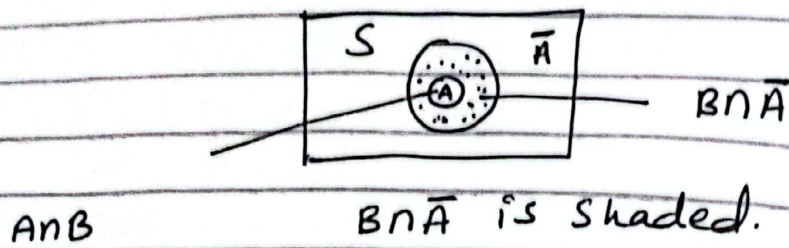
$A \cap B = 0$, A & B are mutually exclusive.

$A \cup B = S$, A & B are exhaustive.

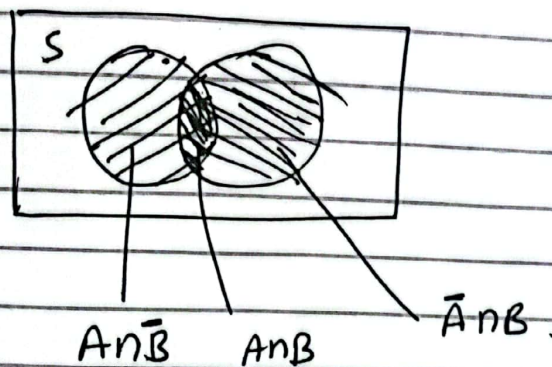
$A \cap \bar{B}$, A occurs but B does not occur

$A|B$, even A occurs given that B has occurred

Venn diagram.



$$\bar{B} = S - B$$



"atleast" and "or" in question

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$- P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

For and in Questions.

~~P(A ∩ B)~~ $P(A \cap B) = P(A) \cdot P(B)$
indep^t A & B.

$$P(A \cap B) = P(A|B) \cdot P(B)$$

or $P(B|A) \cdot P(A)$
For dep^t A & B.

Conditional.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Total prob Law

$$P(A) = P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2) \\ + P(A|B_3) \cdot P(B_3)$$

Bayes' Law

$$P(B_1|A) = \frac{P(A|B_1) \cdot P(B_1)}{P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2)}$$